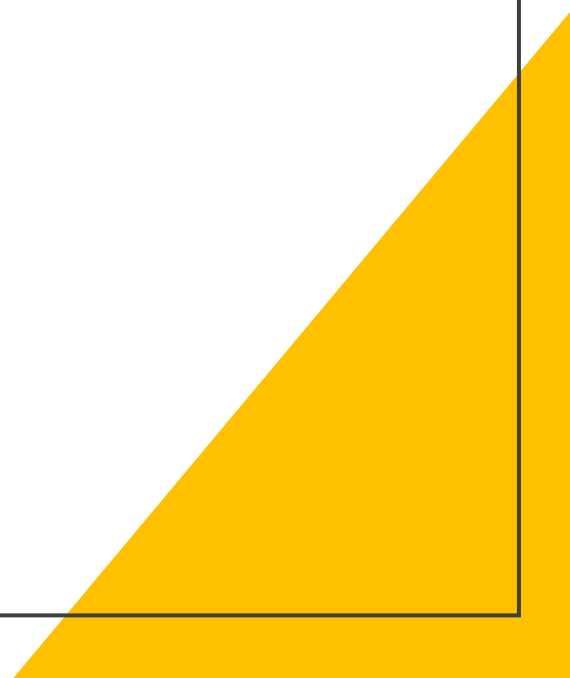


Applied Computational Chemistry

Liviu Ungur
CM5235



Course Plan: Theoretical Lectures

- Overview of 'Computational Chemistry' and its relations to other research areas
- Atomistic and Coarse-Grained Force Field Methods: background and method limitations
- Principles of Quantum Mechanics and Basic Knowledge: operators, eigenvalue equation, quantization, atomic orbitals of H as obtained from SE for H
- Hartree-Fock Theory and Roothaan-Hall equations
- Analysis of the wave function: atomic charges, electronic density, orbital localization, etc.
- Electron correlation methods: MP2, CCSD, etc.
- Molecular dynamics: DFT and semi-empirical calculations

Course Plan: Practical Sessions

- The practical sessions will include straightforward tasks to get familiar with Linux OS, scripting, input and job preparation and submission.
- We will get familiar and use the following QC packages:
Gaussian, ORCA, OpenMOLCAS and VASP.
- **Gaussian** and **ORCA** will be used for computational tasks such as: geometry optimization and evaluation of vibrational frequencies, visualization of orbitals, electronic density, spin density, magnetic exchange (BS-DFT), excited states via TD-DFT method.

Course Plan: Practical Sessions

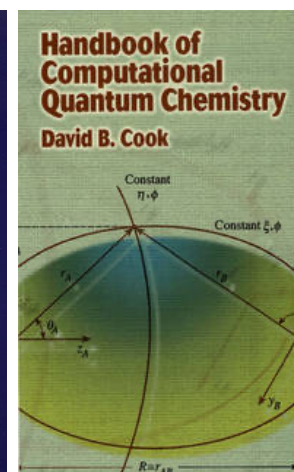
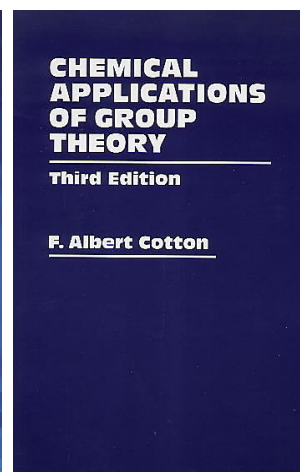
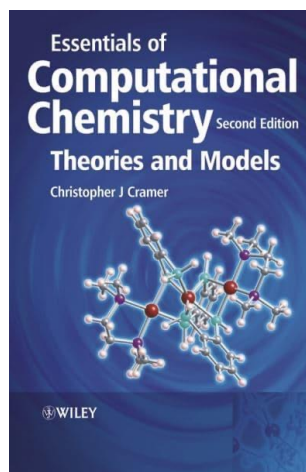
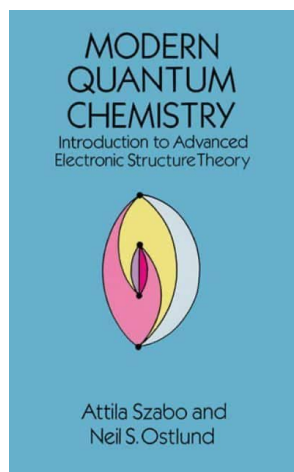
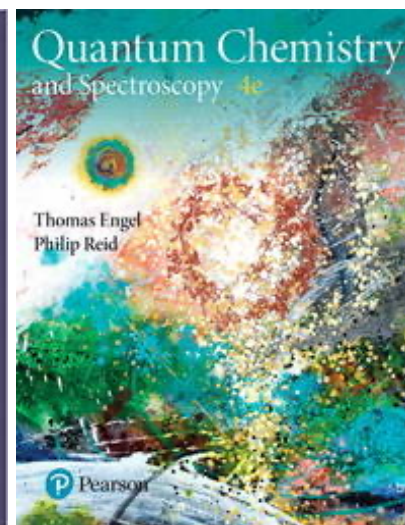
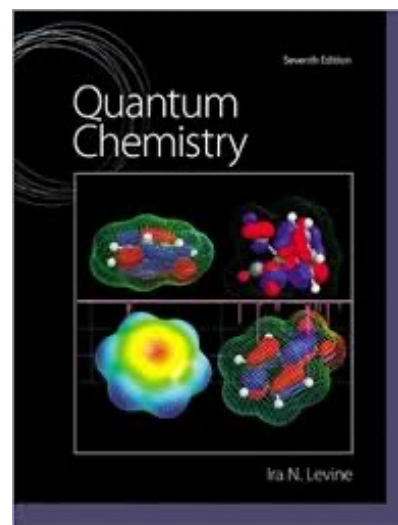
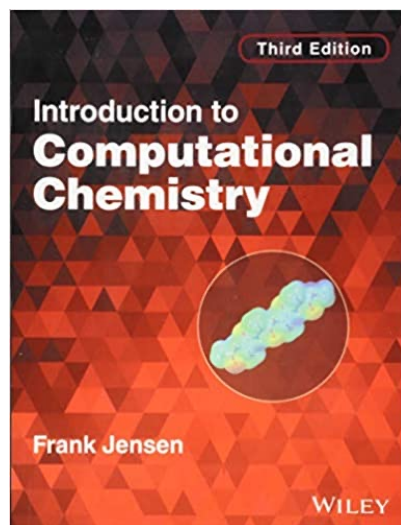
- We will use **OpenMOLCAS** for the multiconfigurational methods RASSCF/CASPT2/RASSI and evaluation of electronic excited states of transition metals and lanthanide compounds
- Finally, **VASP** package will be used for the investigation of solid-state periodic materials, periodic 2D layers and deposition of molecules on surfaces.
- Each of these packages have a wide set of **tools** for processing results (written in *python*, *bash* or else).

Course Plan: Assessments

- Four Homework Assignments, which involve practical computational exercises to be carried out at HPC@NUS cluster
 - GAUSSIAN / GAUSSVIEW (15% weight)
 - ORCA / Avogadro / Chemcraft (15% weight)
 - OpenMOLCAS / Luscus (15% weight)
 - VASP / VESTA / vaspkit / py4vasp (15% weight)
- *Please submit the reports on time, as a single PDF file / student.*
- *Handwritten files are OK – but make sure I can understand your writing!*
- Mid-Term quiz (Tuesday, 29 Sep 7-8 pm) (15% weight)
- Final Exam (25% weight)

The midterm test and final exam will mostly include theoretical and conceptual questions. No practical calculations will be needed for the mid-term quiz and for final exam.

Useful
books to
read



What is Computational Chemistry?

- *Computational Chemistry* is a (relatively new) research area which aims at finding answers to chemical problems by using computers & fundamental laws of interaction between particles
- Example: a typical chemical reaction has many unknowns:



- find the most suitable **reagents**
- find the most suitable **catalyst**
- **reaction conditions** (Temperature, Pressure, solvent, etc.)
- separation of the NEW compound from the reaction mixture
- find its physical properties: molecular and crystal structures, potential energy surfaces, melting point, EPR spectra, UV-VIS, IR, NMR and Raman spectra, etc.
- Establish the **reaction mechanism** allowing to get new understanding / insight
- Understand the **kinetics** of the reaction, allowing to scale it up for industrial application

Important names in this field:

A. Einstein, E. Schrodinger, P. Dirac, V. Fock, J. Pople, W. Kohn, D. Hartree, R. Parr, M. J. S. Dewar, R. Car, M. Parinello, H. F. Schaeffer, A. Warshel, W. Yang, M. Buehl, I. Mayer, R. S. Mulliken, R. Alrichs, P.-O. Löwdin, P. Jørgensen, T. Helgaker, J. Olsen, J. Gauss, E.A. Carter, P. Kollman, M. Levitt, M. S. Gordon, B.O. Ross, F. Neese, H. Lischka, W. Yang, Y. Saad, M. Head-Gordon, W. A. Goddard III, D. P. Craig, N. Allinger, P. Hohenberg, P. Pulay, etc.



Enrico Clementi

Pioneered (1961) ***ab initio quantum chemistry*** applications to small and large molecular systems, with coding and free distribution of Hartree–Fock atomic and molecular programs. First systematic estimate of correlation energy for atoms and molecules. Pioneering (1963) ***density functionals methods*** for atoms and molecules, atomic and molecular relativistic correction, graphics, and since 1972 ***Monte Carlo and Molecular Dynamics*** for liquids, biological systems in aqueous solutions with ab initio potentials. Free distribution of Monte Carlo and Molecular Dynamic codes. Pioneering (1983) ***parallel architecture*** with design and construction of LCAP1, LCAP2 and LCAP-3090 hardware and relative system software with applications to quantum chemistry, molecular dynamics, and micro-dynamics. With G. Corongiu introduction and tests (2006-present) of the Hartree–Fock–Heitler–London method (i.e. ***valence bond theory***) and the Chemical Orbital model.

Important names in this field:



Walter Kohn



John Pople

Nobel Prizes In Computational Chemistry:

- Walter Kohn (1998): "for his development of density-functional theory"
- John Pople (1998): "for his development of computational methods in quantum chemistry"



Martin Karplus, Michael Levitt, and
Arieh Warshel

- Martin Karplus, Michael Levitt, and Arieh Warshel (2013): "for the development of multiscale models for complex chemical systems"

Top 10 most cited computational papers among all-time papers published in JACS:

[ranking]. Title (citations):

- 3.** Development and use of quantum mechanical molecular models. 76. AM1 : a new general purpose quantum mechanical molecular model (14995)
- 6.** A 2nd Generation Force Field for the Simulation of Proteins, Nucleic Acids, and Organic Molecules (10233)
- 10.** Development and Testing of the OPLS All-Atom Force Field on Conformational Energetics and Properties of Organic Liquids (8377)
- 17.** GROUND-STATES OF MOLECULES .38. MNDO METHOD - APPROXIMATIONS AND PARAMETERS (6727)
- 20.** UFF, A FULL PERIODIC-TABLE FORCE-FIELD FOR MOLECULAR MECHANICS AND MOLECULAR-DYNAMICS SIMULATIONS (5977)
- 24.** EFFICIENT IMPLEMENTATION OF THE GAUGE-INDEPENDENT ATOMIC ORBITAL METHOD FOR NMR CHEMICAL-SHIFT CALCULATIONS (5289)
- 26.** A NEW FORCE-FIELD FOR MOLECULAR MECHANICAL SIMULATION OF NUCLEIC-ACIDS AND PROTEINS (4824)
- 28.** SELF-CONSISTENT MOLECULAR-ORBITAL METHODS .21. SMALL SPLIT-VALENCE BASIS-SETS FOR 1ST-ROW ELEMENTS (4517)
- 31.** CONFORMATIONAL-ANALYSIS .130. MM2 - HYDROCARBON FORCE-FIELD UTILIZING V1 AND V2 TORSIONAL TERMS (4257)
- 36.** NATURAL HYBRID ORBITALS (3872)

A Brief History

- 1925
 - Warner Heisenberg, Max Born, and Pascal Jordan develop matrix mechanics
 - Erwin Schrodinger invents wave mechanics and non-relativistic Schrodinger equation
 - Walter Heitler and Fritz London publish first calculations on chemical bonding
- 1927
 - Douglas Hartree publishes self-consistent field method
- 1930
 - Vladimir Fock formulates Hartree-Fock theory
- 1947
 - ENIAC is the first general-purpose computer to be built
- 1950
 - Clemens Roothaan publishes LCAO theory
- 1951
 - UNIVAC is the first commercial general-purpose computer
- 1955
 - First ab initio calculation on 'large' molecule, N₂.
- 1956
 - Transistors replace vacuum tubes in computers
- 1964
 - Pierre Hohenberg and Walter Kohn introduce density-functional theory
- 1970
 - John Pople introduces Gaussian (software)
- 1971
 - First commercially available microprocessor (Intel 4004)
- 1998
 - Isaac Chuang (LANL) and Neil Gershenfeld (MIT) created first quantum processor (using 2- qubit)
- 2019
 - Google Inc. demonstrated **quantum supremacy** using a programmable superconducting quantum processor
- 2020
 - Chinese research team (from USTC) announced a quantum processor using photons

Promising Paths for Future Developments in Computational Chemistry

1) AI / Machine Learning / Deep Learning + Computational Chemistry

- ➔ possibility to obtain DFT-quality results < 10000 times faster compared to common DFT
- ➔ allowing for a more robust scan of the phase space, fast prediction of new materials / phenomena / reactivity / properties
- ➔ accelerating the synthesis and discovery of novel molecules and materials

2) Accurate description of weak interactions: e. g. Van Der Waals, dispersion, long-range correlations

3) Accurate prediction of crystal structures and phonon spectra for molecular crystals

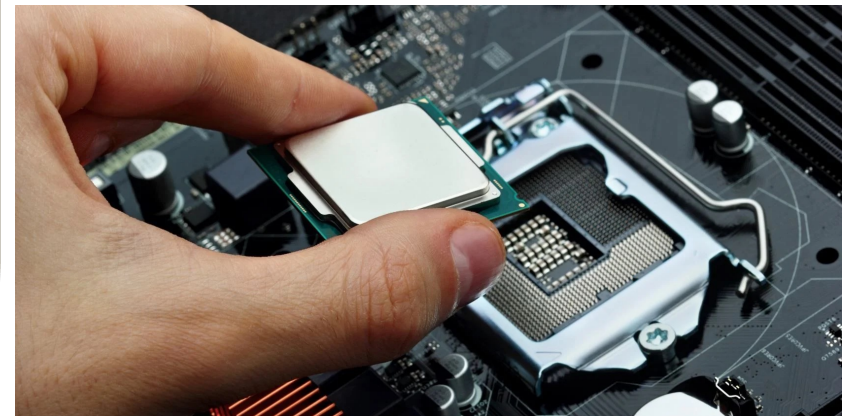
3) Non-equilibrium quantum dynamics, i.e., solving Schrodinger equation for electrons and nuclei concomitantly (non- Born-Oppenheimer)

4) Development of computational methods for Quantum Processor and various hybrid methods

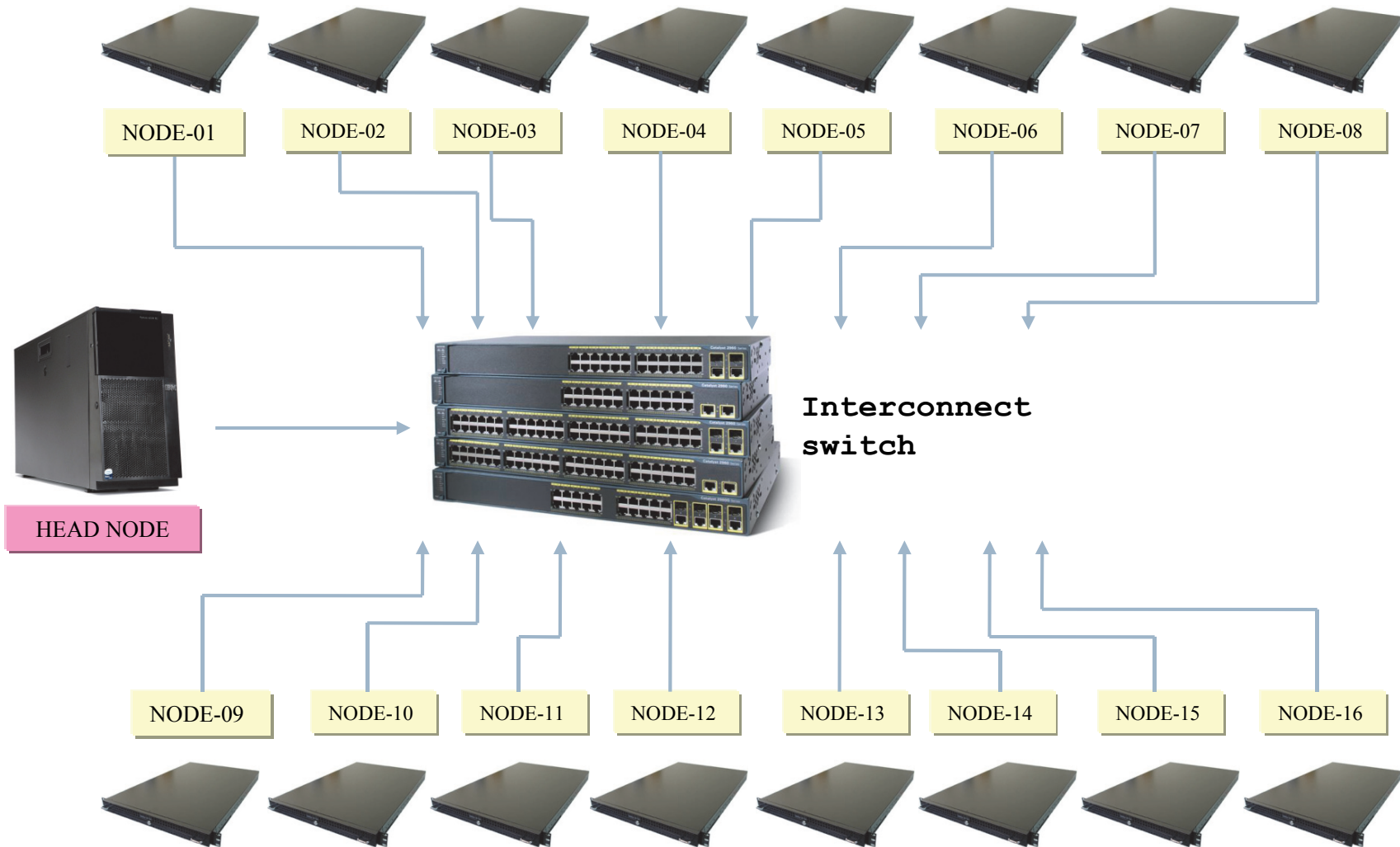
Brief description of computers,
supercomputers and their performance

The Anatomy of a Computer

- A **mainboard** which integrates the following components:
- **CPU(s)** containing
 - several computing cores
 - local cache memory (very fast)
- Active **RAM** Memory (fast access)
- Hard disc memory (slow access)
- Graphics Card (containing its own GPU and active memory)
- Peripheral components



The Anatomy of a Computer Cluster



The Anatomy of a Computer Cluster (e.g. NSCC)

- Heterogeneous interconnect – slow fast data transfer between the processors (cores) on tdata transfer between nodes and he same node
- Quite cheap -- the price of the cluster is only slightly larger than the sum of prices of all nodes. Large deployments are possible
- Fault tolerant: if one node fails to respond – it can be switched off alone. No need to shut down the cluster



The Anatomy of a Computer Cluster

- Heterogeneous interconnect – slow data transfer between nodes and fast data transfer between the processors (cores) on the same node
- Quite cheap -- the price of the cluster is only slightly larger than the sum of prices of all nodes. Large deployments are possible
- Fault tolerant: if one node fails to respond – it can be switched off alone. No need to shut down the cluster
- Usually run on LINUX OS



Shared-memory supercomputer (e.g. IBM Blue Gene/Q)



IBM Blue Gene/Q at Argonne National Lab

Commercial supercomputers (e.g. Blue Gene/Q) have proprietary large-scale motherboard which is capable to host and integrate efficiently a large number of processors and RAM memory slots. The price for such supercomputers is much higher than usual computer clusters with equivalent number of nodes/cores.

The very large RAM memory and availability of thousands of nodes allows unique tasks which cannot be done elsewhere: large matrix diagonalisation.

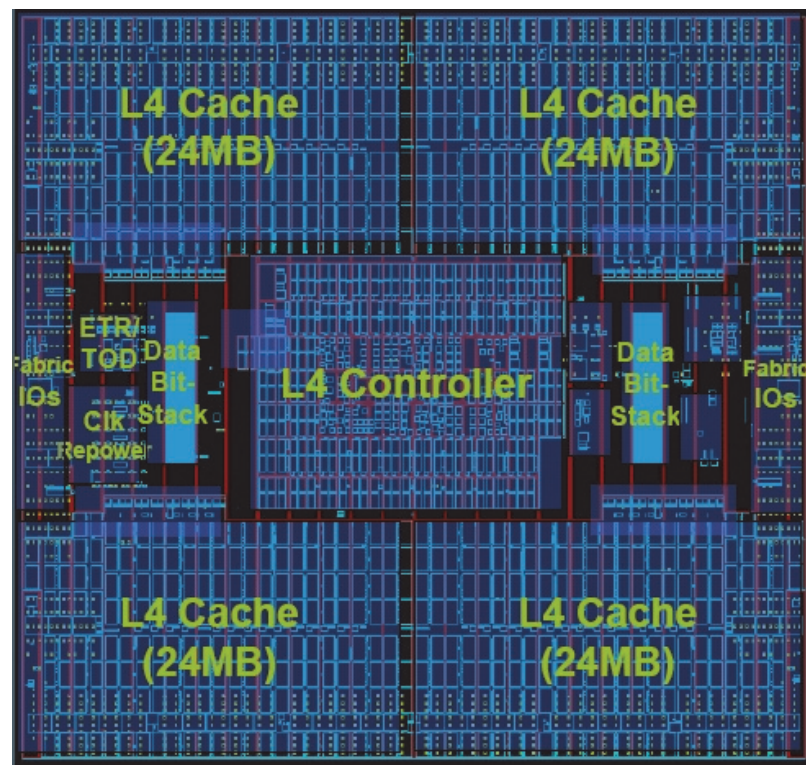
They run on UNIX OS with a set of in-house optimised tools.

Memory bandwidth and latency

Latency: is the time it takes for the data requested by the CPU to start arriving

Bandwidth: is the speed (rate) of the arriving data

Location	Latency	Bandwidth
CPU registers	1 clock	-
L1 cache	1 clock	150 GB/s
L2 cache	3 clocks	50 GB/s
L3/L4 cache	10 clocks	50 GB/s
Own RAM	100 clocks	10 GB/s
NUMA RAM	150 clocks	10 GB/s
Other node RAM	10k clocks	1 GB/s
Flash storage	10k clocks	1 GB/s
Magnetic storage	10k clocks	0.1 GB/s



Smaller datasets and programs are processed faster because they fit into a higher level cache memory (which works faster)
Random hits to the main RAM can make the program very slow.

Memory bandwidth and latency

The two often used performance characteristics of a CPU are:

GIPS (billion instructions per second) – how many machine code instructions a processor can execute per second. This metric is very architecture- specific.

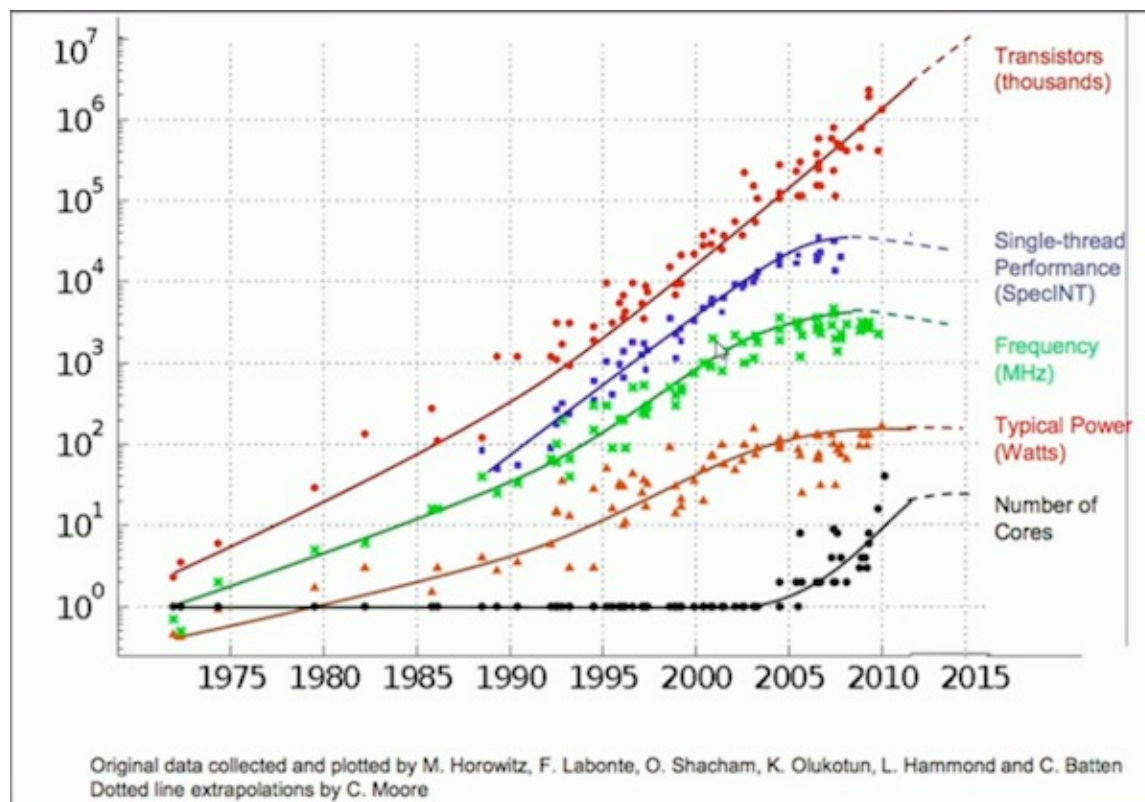
GFLOPS (billion floating-point operations per second) – how many single- or double-precision multiplications a processor can perform per second. Scientific calculations are nearly always performed in double precision.

Power consumption is not quite crucial for a single computer / node, on average 0.2 kWh. Energy consumption is a problem for large computer clusters.

Novel ARM processors (used in mobile phones and tablets) use significantly less energy compared to INTEL's processors (as example the new Apple M1 processors).

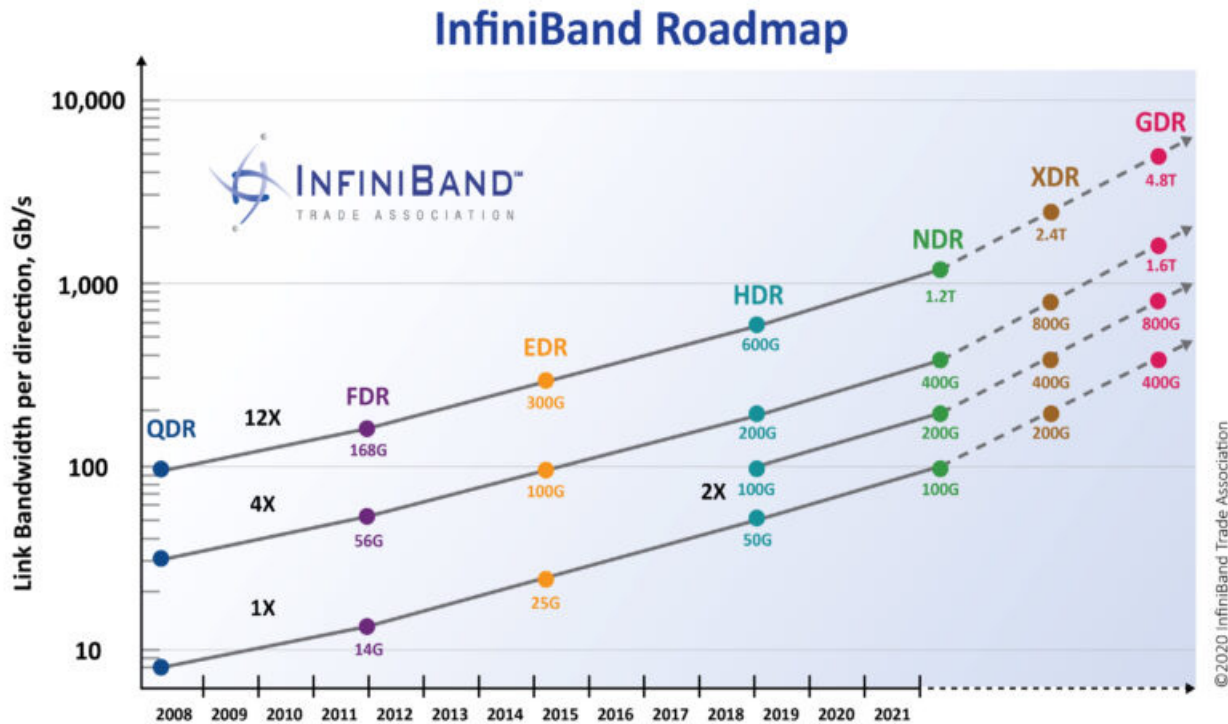
Current TOP Supercomputer, namely Japan's **Fugaku** uses ARM-based processors A64FX.

The Moore's Law



The number of transistors on integrated circuits doubles approximately every two years. The ever-increasing computing power of modern computers drives the extension of computational chemistry to more complicated systems with more sophisticated levels of theory

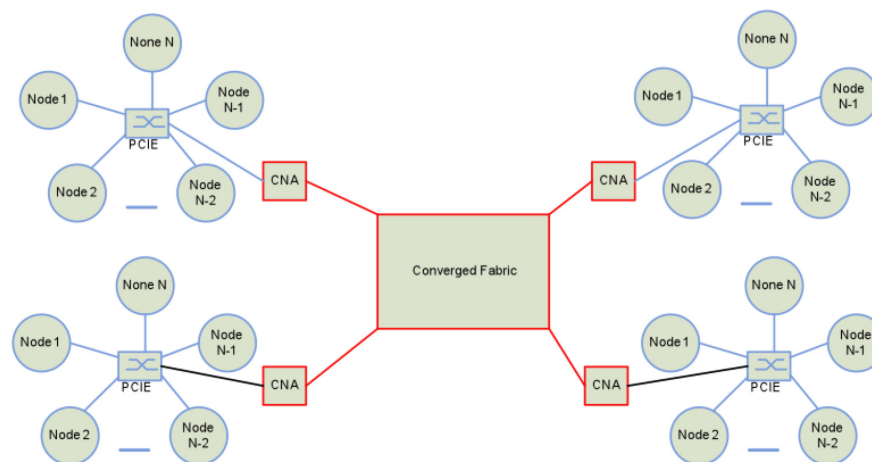
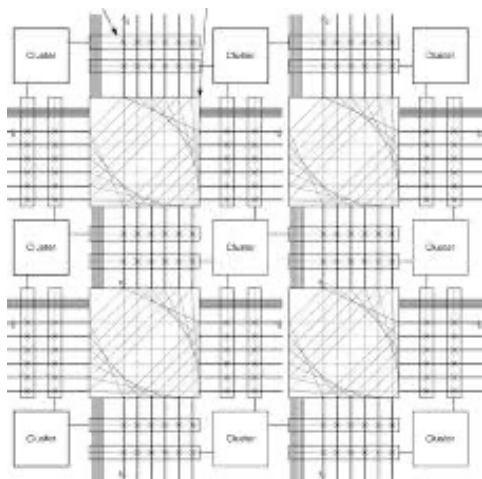
Network interconnects



Name and Year	Bandwidth
10BASE Ethernet (1990)	10 Mbps
100BASE Ethernet (1995)	100 Mbps
1000BASE Ethernet (1999)	1 Gbps
Myrinet-10G (2005)	10 Gbps
10GBASE Ethernet (2006)	10 Gbit/s
Infiniband 12X QDR (2009)	96 Gbit/s
100GBASE Ethernet (2010)	100 Gbit/s

The protocols may be implemented over copper cable or optical fibre. Noise and impedance losses mean that at larger bitrates the maximum length of the cable gets shorter. Latencies are typically in the 100 ns range.

Cluster Network interconnect

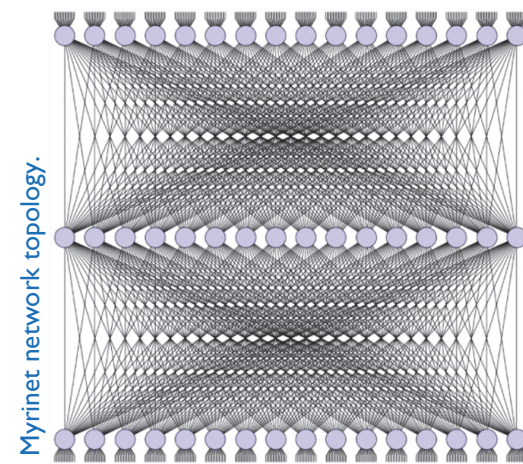


The problem: assume that we have distributed a given computational problem among N different nodes, where N is large. At a certain point in the calculation some data needs to be exchanged or shared among certain nodes. How to efficiently share/exchange the necessary data only among the nodes which need it?

➔ Design of an efficient interconnector allowing to share data between nodes is a highly non-trivial problem. Most of the top-notch solutions are guarded by commercial patents and specialised software.

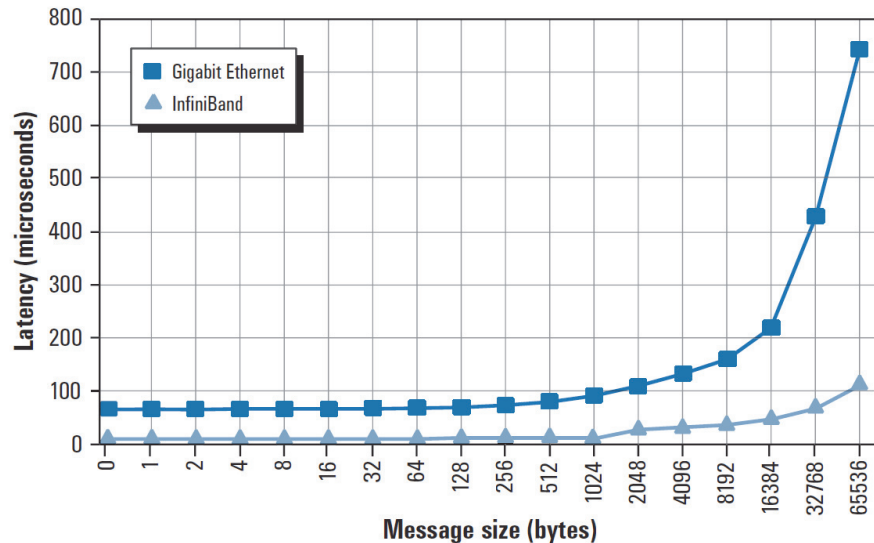


256-port Myrinet switch.



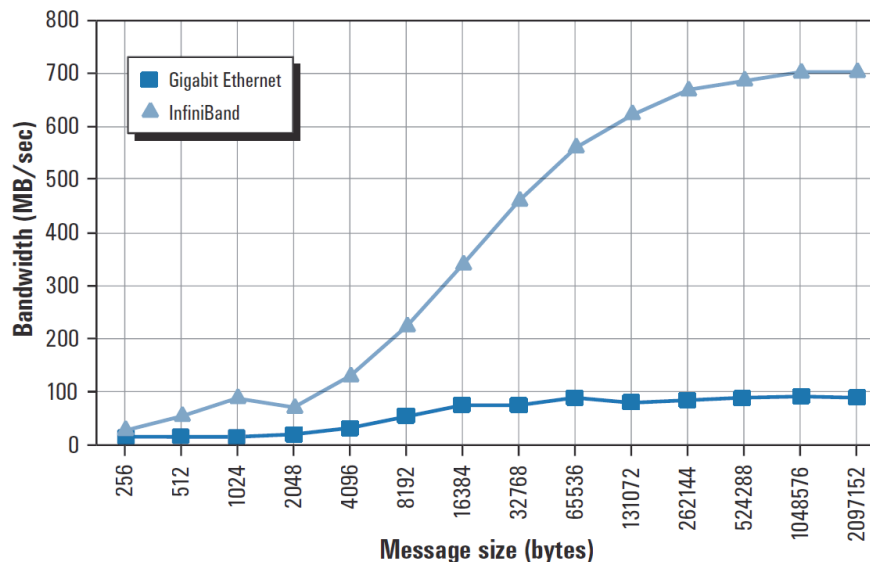
Myrinet network topology.

Cluster bandwidth and latency



Note the microsecond latencies – 1 microsecond is equivalent to about 2k clocks on a Nehalem CPU. The bandwidth is also far below the RAM or cache bandwidth.

For this reason, memory intensive calculations with a lot of data dependencies are often difficult to run on cluster architectures. Running them on a single node could be much faster than in parallel on several nodes.



Filesystem performance

Name	Latency	Bandwidth	Typical usage
RAM disk	nanoseconds	10 GB/s	A segment of RAM that looks like a file system to the OS, not often used.
scratch / swap	microseconds to milliseconds	0.1-1 GB/s	Dedicated filesystem, usually very fast, for the storage of intermediate calculation results.
local filesystem	milliseconds	0.1 GB/s	Stores operating system files, executables, system logs, etc.
network filesystem	milliseconds	0.1 GB/s	User home directories located on some external hardware that is accessed via a network.
backup filesystem	seconds	0.1 GB/s	Infrequently accessed incremental backup areas.

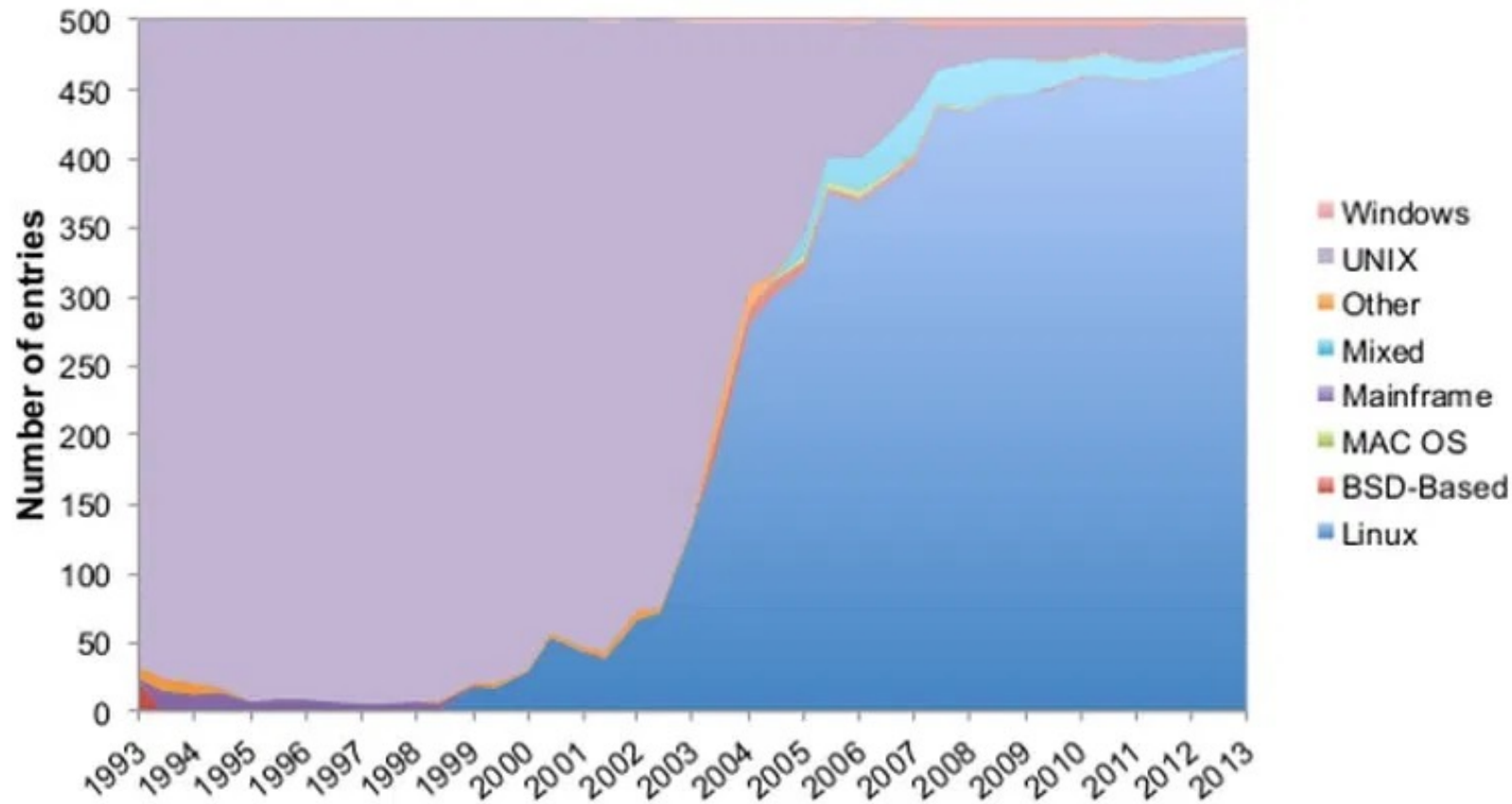
Large Input-Output (IO) operations tend to slow down the cluster performance significantly. Use the 'scratch' folder for running the calculations, as indicated by cluster admin.

Scaling of elementary operations (BLAS & LAPACK):

Operation	CPU Cost / FLOPs	Memory cost	Sparse cost	Parallel execution
Matrix-matrix multiplication	$O(n^3)$	$O(nnz)$	LOWER	EASY
Matrix-vector multiplication	$O(n^2)$	$O(nnz)$	LOWER	EASY
exp(matrix)-vector multiplication	$O(n^2)$	$O(nnz)$	LOWER	EASY
Matrix exponentiation	$O(n^3)$	$O(nnz)$	LOWER	EASY
Most matrix factorizations	$O(n^3)$	$O(n^3)$	SAME	HARD

Matrix factorisations (SVD, LU, diagonalisation, Householder, etc.) are very expensive, difficult to parallelize, and do not benefit from sparse storage.

LINUX dominates the OS of top supercomputers



Since 2018, ALL of the top 500 supercomputers are running on Linux OS

Hands-on Practical Session #1

Task 1: Download and install **NUS** VPN software on your laptop and login to VPN of the NUS. Check here:

<https://nusit.nus.edu.sg/eguides/>

Task 2: Request an account for HPC@NUS. Click on the link and register for an account.

<https://nusit.nus.edu.sg/hpc/get-an-hpc-account/>

After this is done ➡ try to log in!

Hands-on Practical Session #1: SSH Login

Windows: You can use “**Putty**” -- a client to connect to the HPC.

<https://www.chiark.greenend.org.uk/~sgtatham/putty/latest.html>

Mac OSX: Just open the **Terminal** app and use it to connect to various servers via the **ssh** command.

Task 3: Connect via ssh to your account on HPC server of NUS.

Apply for HPC account

<https://nusit.nus.edu.sg/hpc/get-an-hpc-account/>



Information
Technology

HPC@NUS IT

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Register for HPC

Instructions for registering an account

- Please Log into NUS VPN (if you are outside of campus)
- Please use **Chrome**, **Edge** or **Firefox** browsers only.
- Click on the Registration button below to start registering

[Click to Register](#)

Introductory Notes:

Please read and practice!!!

Mandatory reading:

<https://nusit.nus.edu.sg/services/getting-started/introductory-guide-for-new-hpc-users/>

How to run batch jobs in HPC:

<https://nusit.nus.edu.sg/services/getting-started/how-to-run-batch-job/>

Hands-on Practical Session #1

Task 5: Windows users may install **Gaussian** and **GaussView** software on their laptops, after they sign their agreement to the terms & conditions (document sent separately).

Mac OSX Users may access this software as well, but they need to have a Windows emulator software (e.g. Boot Camp, Parallels Desktop, Virtual Box, VMWare Fusion, etc.).

I will share the Windows-64 installation files for the above-mentioned software only to those who send the signed agreement to my email.

Gaussian and **GaussView** can be used at HPC@NUS server without any installation (as they are already available there for all users)

Hands-on Practical Session #1

ORCA is a closed-source, free-to-use-for-academia quantum chemistry package developed at the Max Planck Society, Germany. It can also be installed natively on Windows, but it lacks a graphical user interface and can only be used from the command prompt.

<https://orcaforum.kofo.mpg.de/filebase/>

(registration for a user account on ORCA Forum is required. It is free of charge. Download for academic usage is free.)

ORCA 6 is available as a module in HPC@NUS (no installation required).

Hands-on Practical Session #1

OpenMOLCAS <https://gitlab.com/Molcas/OpenMolcas> (free and open-source software)

Pre-compiled executables and running scripts are provided via an installation script (to be sent separately via Canvas).

VASP <https://www.vasp.at>

(Commercial and closed software)

Pre-compiled executables and running scripts are provided via an installation script (to be sent separately). Additional python post-processing scripts for VASP are free of charge and can be installed freely, either on your laptop or under your account in the HPC server

Practical Session #1

Chemcraft: <http://www.chemcraftprog.com/index.html>

(visualization and interpretation of computational results: it is a commercial software, but the trial period of 150 days!)

VMD: <https://www.ks.uiuc.edu/Research/vmd/>

(FREE, visualization tool for molecular dynamics and large biomolecules)

Multiwfn: <http://sobereva.com/multiwfn/>

(FREE, visualization and computation of many useful properties, using the computed wave function)

Visualization Software

Avogadro: <https://avogadro.cc>

(FREE, visualization and editing of molecules)

JMOL: <http://jmol.sourceforge.net>

(FREE, visualization and editing of molecules)

MOLDEN: <https://www3.cmbi.umcn.nl/molden/>

(FREE, visualization and computation of many useful properties, using the computed wave function)

Material Studio: <https://nusit.nus.edu.sg/services/hpc/application-software/material-studio/>

(commercial software for computational modelling of materials). Free to use at NUS, although the number of available licenses is not large.

Useful software

PUTTY: <https://www.putty.org>

(FREE, used for SSH connection to HPC clusters)

FileZilla: <https://filezilla-project.org>

(FREE, used for transferring files to and from HPC cluster)

WinSCP: <https://winscp.net/eng/download.php>

(FREE, Windows, used for transferring files to and from HPC cluster)

Notepad++: <https://notepad-plus-plus.org/downloads/>

(FREE, Windows, used for viewing and editing text files)

Useful reading:

<https://nusit.nus.edu.sg/services/getting-started/introductory-guide-for-new-hpc-users/>

Useful reading

Useful reading:

- The Linux Directory Structure, Explained

<https://www.howtogeek.com/117435/htg-explains-the-linux-directory-structure-explained/>

- How to run a script in Linux

<https://www.cyberciti.biz/faq/howto-run-a-script-in-linux/>

- Introduction to HPC@NUS:

<https://nusit.nus.edu.sg/services/getting-started/introductory-guide-for-new-hpc-users/>

- Useful websites to learn bash, python and other coding languages:

<https://www.hackerrank.com>

Further tasks:

- 1) Open the **Linux Terminal** (e.g. Ubuntu, or Terminal app on OSX) and perform the following small tasks (only using the command line in the terminal):
 - a. create a new text file and fill some random text inside (about 1000 words, 30+ lines of text, copy paste from internet). You may use the **vi** or **nano** command.
 - b. create a new folder and move the file into the folder (use **mkdir** and **mv** commands)
 - c. make a duplicate of this file (use **cp** command)
 - d. modify the duplicated file to remove one word from it (use **vi** or **nano**).
 - e. compare the two files (use **diff** command)
 - f. calculate how many times the word “**and**” is present in the file (use the three commands **cat** and **grep** and **wc** with concatenation). e.g.:
cat file1 | grep 'and' | wc -l

Further tasks:

2) continuation:

- a. transfer the two files to the HPC server (use **scp** command)
scp file1 user@server:~/
- b. connect to the HPC server using the command line (use **ssh** command)
ssh user@server
- c. create and modify a new file (file3) in your \$HOME on the HPC server, similar to task (a. and b.)
- d. transfer the file3 to your own laptop, using **scp** command or FileZilla.
- e. Read about job submission (#PBS system) and follow (execute, repeat) the existing examples: <https://nusit.nus.edu.sg/services/getting-started/how-to-run-batch-job/>

Further tasks:

3) continuation:

- c. write a simple script which generates a text file. Example below:
- d. submit it and check if file4 was generated.

```
#!/bin/bash
#PBS -P test_script
#PBS -q parallel
#PBS -l select=1:ncpus=12:mem=2GB
#PBS -j oe

cd $PBS_O_WORKDIR
if [ -f file4 ]; then rm -rf file4; fi

echo 'some random text ' > file4
echo 'PBS_O_WORKDIR    =' $PBS_O_WORKDIR >> file4
echo 'ncpus            =' $ncpus >> file4

exit
```

Homework (this is not an assignment):

Get familiar as much as possible with:

- *command line* (vi, ssh, scp, qsub, cat, sed, awk, bash scripting, regular expressions, etc.) The more you use them, the easier your life becomes inside the terminal
- *Linux system* (environment variables, \$PATH, \$LD_LIBRARY_PATH, file line endings in Linux and Windows)
- *Submission of jobs* on HPC and NSCC (they are similar but there are some minor differences to take care of.
- *Transfer of files* to and from each of these servers and your laptop.